Rithvik Vadlamani

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EDUCATION

The University of Texas at Austin B.S. in Electrical and Computer Engineering May 2028
Coursework: Software Design and Implementation I and II, Computer Architecture, Data Structures and Algorithms, Operating Systems

TECHNICAL SKILLS

Languages & Backend: Python, Java, C, SQL, FastAPI, JavaFX, TCP sockets, Gson, multithreading, synchronization, REST APIs
AI & Data: PostgreSQL, SQLite, NumPy, ONNX, vector search, retrieval-augmented generation (RAG), OpenAI/Ollama APIs
Tools & Testing: Git, Linux, GitHub Actions, pytest, JUnit 5, API testing, OpenAI Codex, KiCad CLI, Markdown, YAML, XML

PROJECTS

Internship Alert System | Python, FastAPI, PostgreSQL, ONNX, GitHub Actions Jul. 2026 to Aug. 2026

- Built a personal internship alert tool to reduce manual searches, missing deadlines, and be informed earlier about newly posted roles.
- Developed a FastAPI dashboard backed by PostgreSQL for 40,000+ job records and counting, adding custom filters, password protection, and row-level security across ten tables to organize relevant openings and organize application data for personal use.
- Implemented an eight-thread job collection pipeline with URL deduplication, retries, and progressive database commits to retain progress after failures, checking 100+ company boards and finding 89 new eligible roles in one measured 284-second collection run.
- Introduced an alert system to notify myself of new roles matching my filters through email and ntfy mobile push notifications.
- Integrated ONNX with keyword scoring and cached vectors to rank jobs against my own resumes, to create a ranking of the jobs.
- Created tracking for five application stages, recording applied roles, dates, notes and excluding completed items from discovery.

LC-3b Processor Simulation & Virtual Memory | C Apr. 2026

- Built a C simulator for an LC-3b processor with five pipeline stages, modeling control and data latches and instruction cache latency over multiple cycles while implementing stalls without forwarding and instruction squashing for branches, jumps, and traps.
- Validated pipeline timing and final register and memory state against reference simulators using assembly tests for consecutive dependencies, dependencies after loads, and branch hazards, comparing stall and squash counts with expected cycle behavior.
- Extended a separate microcoded LC-3b simulator with virtual memory by adding address translation states and widening microinstructions for new control signals, handling page faults and other exceptions with correct restart and handler vectoring.

Multithreaded Library Management System | Java, JavaFX, TCP sockets, Gson, JUnit 5 Apr. 2026

- Built a Java library application with a JavaFX client and TCP server using Gson to exchange JSON requests, keeping the interface responsive by performing socket communication on background threads and updating JavaFX controls through Platform.runLater().
- Implemented a thread for each client connection and synchronized borrowing, return, and hold operations to prevent concurrent borrowers from receiving the same copy, with an ordered hold queue that reserves returned items for the next waiting member.
- Tested login, duplicate username rejection, and concurrent borrowing with JUnit 5 and filesystem fixtures isolated by @TempDir.

KiCad Datasheet RAG Assistant | Python, SQLite, NumPy, OpenAI/Ollama APIs, pytest Mar. 2026

- Built a Python retrieval-augmented generation pipeline combining PDF extraction, keyword and vector search, and KiCad connectivity graphs to support component questions with page-level citations using configurable local and cloud models.
- Implemented incremental indexing, project isolation, and source-quote validation to reject stale indexes and invalid citations, with 38 automated tests covering retrieval, netlist parsing, electrical calculations, and local and cloud API request and error handling.

Codex Skill for PCB Automation | OpenAI Codex, Markdown, KiCad CLI Mar. 2026

- Developed a reusable Codex skill that guides AI agents from electronics requirements through schematic generation using modular Markdown instructions, reference files, and YAML metadata for automatic discovery and selective loading of task-specific guidance.
- Defined a KiCad workflow with version-aware CLI commands, explicit design constraints, and verification criteria, combining electrical rule checks, netlist comparisons, and rendered-output review to check intended connections and schematic readability.

EXPERIENCE

Guadaloop | Power Electronics Team Jan. 2026 to Present

- Designed Guadaloop power PCBs with buck converters and LDOs to step down a 60 V battery for onboard electronic subsystems requiring lower voltages and distribute 60 V directly to the levitation and fluids systems according to their power requirements.
- Assembled custom circuitry and an inverter to supply Guadaloop's propulsion system from the pod's 600 V battery source.
- Modeled a faraday cage enclosure to shield the pod's electronics and limit electromagnetic interference with nearby subsystems.
- Collaborated with members to work on the existing low voltage board to solder SMD components to mitigate with high frequencies
- Co-designed a four-layer, 60 V H-bridge PCB for magnetic levitation, integrating gate drivers, and bidirectional current sensing.